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Phylogeography of *Microphis retzii* (Bleeker, 1856) and *Microphis brachyurus* (Bleeker, 1854) in the Pacific

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Abstract

Syngnathidae are a charismatic family of teleost fishes, represented by seahorses, seadragons, and pipefishes. Syngnathidae are mainly composed of marine species, but about 30 species of pipefishes inhabit freshwater insular environments of the Indo-Pacific realm. Recent research has shown that some freshwater pipefish species are amphidromous and exhibit high intraspecific divergences across their distribution range, like *Microphis brachyurus* (Bleeker, 1854) distributed from Sri Lanka to French Polynesia and *Microphis retzii* (Bleeker, 1856) distributed from Taiwan to Indonesia. In this study, we used the mitochondrial cytochrome oxidase I (*COI*) partial gene of 91 specimens of *M. brachyurus* and 30 specimens of *M. retzii* from localities representative of their respective distribution area to improve knowledge on the population structure of these two widespread species. Genetic species delimitation and phylogeographic analyses were combined to explore spatial patterns of genetic diversity across the distribution ranges of the two species. We have highlighted deep genetic structuring within the two species and relate these results to various biotic and abiotic factors. For *M. brachyurus*, the population in Polynesia is distinct from those in the West Pacific, suggesting its distinctiveness and recognition as an evolutionary significant unit (ESU). For *M. retzii*, three lineages are delimited in its range distribution, suggesting the existence of two distinct species in Southeast Asia (Bali/Java/Lombok and China/Taiwan). Pipefish species are particularly vulnerable to anthropogenic pressures (inherent to Syngnathidae and insular environments). The present results, revising species delimitation and geographic distribution, will help implement effective conservation and management measures.

KEYWORDS

amphidromy, *COI*, conservation, dispersal, freshwater pipefish, sequence-based species delimitation, Syngnathidae

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1 | INTRODUCTION

The Indo-Pacific realm is composed of many islands (Briggs, 1974), and it is considered as the largest biogeographic region in the world with several biodiversity hotspots (Mittermeier et al., 2011). Islands of the Indo-Pacific are characterized by freshwater ecosystems that are subject to significant climatic and hydrological fluctuations through time, such as droughts, cyclones, or flood episodes (Gordon, 2003). These fluctuating insular environments have only been colonized by species having developed a life cycle adapted to these conditions: diadromy (Myers, 1949). Diadromous species migrate between seawater and freshwater biomes (McDowall, 1992). In tropical insular systems, amphidromy, a type of diadromy for which migrations are not conducted for reproduction purposes unlike anadromy or catadromy, is the most widespread life cycle (Keith et al., 2015; Keith & Lord, 2011). This particular strategy has evolved in multiple teleosts, crustaceans, or gastropods taxa. These amphidromous lineages contribute to most of the biodiversity occurring in the fresh waters of the Indo-Pacific, with usually high levels of endemism (Nelson et al., 1997). Amphidromous species lay their eggs in rivers (McDowall, 2007); after hatching, juveniles are carried by the current to the sea (Ellien et al., 2011) where they undergo a planktonic phase and stay from 1 to 6 months depending on the species (Lord et al., 2010; Mennesson et al., 2015; Tabouret et al., 2014). After this marine phase, juveniles return to fresh water to colonize rivers in which they grow and reproduce (Keith et al., 2008). The marine phase of amphidromous species is important for their dispersal; it will likely influence connectivity among populations (Cowen & Sponaugle, 2009) and plays a key eco-evolutionary role at the population level (Levin, 2006). The marine dispersal phase, in conjunction with the characteristics of the freshwater landscapes of Indo-Pacific insular systems, influences the spatial structuration of amphidromous populations and species at local (rivers, island, archipelago) and regional scales (Jamonneau et al., 2024; Keith, 2003). The direct observation of marine dispersal is no easy task as there are many factors influencing dispersal, such as ocean currents, larval behavior (e.g., swimming depth), length of marine phase, or the presence of barriers (Lord et al., 2010). It is currently challenging for researchers to study the marine dispersal of amphidromous species directly. Consequently, indirect approaches must be employed to enhance our knowledge of migration paths and dynamics of these species.

Syngnathidae are a charismatic group of teleosts represented by seahorses, seadragons, and pipefishes. They are well known for their specific parental care, as males incubate the eggs in a specialized ventral structure, the brood pouch. Although mainly marine species, about 30 species of pipefishes inhabit insular freshwater systems (Dawson, 1985; Nelson et al., 2016). This group is mainly found in the tropical island environments of the Indo-Pacific area, with only two species currently observed in the Atlantic tropical area: *Microphis lineatus* (Kaup, 1856), from the Caribbean to Brazil, and *Microphis aculeatus* (Kaup, 1856), on the coasts of East Africa. Freshwater pipefish have not been studied for over 35 years, and their taxonomy and nomenclature are unclear and solely based on morphological characters (Dawson, 1984), which have led to confusion regarding genera and species delineations. However, Haÿ et al. (2023a), in their taxonomic revision of the freshwater pipefishes, highlighted significant intraspecific variations, linked to distinct geographical areas, in four mitochondrial genes for two widely distributed species: *Microphis brachyurus* (Bleeker, 1854), distributed from Sri Lanka to French Polynesia, and *Microphis retzii* (Bleeker, 1856), distributed in Taiwan, Philippines, and Indonesia. Moreover, using otolith microchemical analysis, Haÿ, Berland, et al. (2023b) have shown that freshwater pipefishes are amphidromous; these species have a marine dispersal phase, which provides them the potential to colonize distant and fragmentated habitats. The purpose of the present work is to improve our understanding of phylogeographic patterns in these two species and further explore distribution patterns based on mitochondrial lineages.

Understanding variations in fish species composition among different areas is a crucial step toward effective management and preservation of biodiversity (Kong et al., 2017). Despite the fact that insular freshwater systems of the Indo-Pacific have flourishing biodiversity, they are under serious threats due to deforestation, water pollution, and habitat fragmentation caused by damming for irrigation and hydroelectric power stations (Hubert et al., 2015; Kadarusman, Hubert, et al., 2012). The knowledge of population genetic structuring and connectivity among populations is paramount for conservation purposes (Van der Meer et al., 2015). By examining gene in space and time, phylogeography has the ability to identify non-recognized biodiversity units, populations unique to certain geographic areas, and provide insights on speciation (Avise, 2000). The simultaneous study of multiple species in the same geographic area can help identify congruent patterns in the distribution of genetic lineages, thus allowing the definition of evolutionary significant units (ESUs) (Hutama et al., 2017; Moritz, 1994). These units provide priority guidelines for conservation plans (Moritz & Faith, 1998). Many studies focused on the genetic structure and distribution patterns of populations in the Indo-Pacific Ocean (Keyse et al., 2014), but mainly on marine organisms and few on amphidromous species (Abdou et al., 2019; Castelin et al., 2013; Chabarria & Pezold, 2013; Jamonneau et al., 2024; Lord et al., 2012; Mennesson et al., 2018; Taillebois et al., 2013). These studies highlighted varying degrees of spatial genetic structuring between the different geographic zones, and evidenced the presence of past and present biogeographic barriers that limited marine dispersal.

Syngnathidae face multiple pressures linked to their use in traditional medicine, aquarium trade, or sold dried as gifts (Vincent et al., 2011). Although that is particularly true for seahorses, pipefishes are also affected by these threats (Vincent et al., 2011). In the case of freshwater pipefish, the most significant pressures arise from habitat loss and degradation due to changes in natural systems and pollution, which are exacerbated by the fragmented nature of freshwater habitats, making them one of the most endangered groups within the Syngnathidae family (Pollom et al., 2021). Improved knowledge of the ecology of *M. brachyurus* and *M. retzii* will provide key information to local and international managers and stakeholders for the implementation of conservation measures. The aim of the present study is to investigate the mitochondrial genetic diversity and phylogeographic patterns of two freshwater pipefish, *M. brachyurus* and *M. retzii*, across their distribution range. To do so, cytochrome oxidase I (*COI*) sequences were used to clarify the genetic boundaries and distribution of these two species.

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Phylogeographic analyses were conducted to explore the spatial distribution of mitochondrial genetic diversity, and species delimitation algorithms were used to delimit molecular operational taxonomic units (MOTUs), which may require a subsequent reevaluation of their biological status. Species distributions and phylogeographic patterns were then interpreted in the context of current and past aquatic landscapes.

2 | MATERIALS AND METHODS

2.1 | Sampling

In the present study, a total of 91 individuals of *M. brachyurus* and 25 individuals of *M. retzii* were sampled using a DEKA 3000 electrofishing system (Gerätebau, Marsberg, Germany). Specimens were euthanized by an overdose of clove essential oil, following Annex IV of Directive 2010/63/EU. All samples were preserved in 95% ethanol for molecular analysis. In addition to these specimens, we added five sequences of *M. retzii* from China and Taiwan from the online database GenBank. Sampling is summarized in Table 1 and Figure 1.

2.2 | Ethics statement

For Indonesian specimens, the present study and all associated methods were carried out in accordance with the relevant guidelines and regulation of the Indonesian Ministry of Research and Technology. Field sampling in Indonesia was conducted according to the research permits 7/TKPIPA/FRP/SM/VII/2012, 68/EXT/SIP/FRP/SM/VIII/2013, 41/EXT/SIP/FRP/SM/VIII/2014, 361/SIP/FRP/E5/Dit.KI/IX/2015, 50/EXT/SIP/FRP/E5/Dit.KI/IX/2016, 45/EXT/SIP/FRP/E5/Dit.KI/VIII/2017, and 392/SIP/FRP/E5/Dit.KI/XI/2018 for N. Hubert. The other specimens were sourced from the collection at the National Museum of Natural History of Paris.

2.3 | PCR amplification and DNA sequencing

For the specimens fixed in ethanol, DNA was extracted from caudal-fin clips using the Macherey and Nagel NucleoSpin 96 Tissues kit following the manufacturer's instructions on an Eppendorf epMotion 5075 robot.

A mitochondrial fragment of the *COI* gene (650 base pair [bp]) was amplified using the tailed fish specific primers VF2-t1 5′-TGTAAAACGACGGCCAGTCAACCAACCACAGACATTGGCAC-3′; FishF2-t1 5′-TGTAAAACGACGGCCAGTCGACTAATCATAAAGATATCGGCAC-3′; Fishr2-t1 5′-CAGGAAACAGCTATGACACTTCAGGGTGACCGAAGAATCAGA-3′ (Ward et al., 2005); Fr1d-t1 5′-CAGGAAACAGCTATGACACCACAGGGTGTCCGARAAYCARAA-3′ (Ivanova et al., 2007). DNA was amplified using PCR in a final volume of 20 μL containing 2 μL of buffer, 1 μL of dimethyl sulfoxide (DMSO), 1 μL of bovine albumin serum (BAS), 0.8 μL of deoxynucleotide triphosphates, 0.32 μL of each forward and reverse primer, 0.06 μL of Taq DNA polymerase (Qiagen), 2 μL of DNA, and water. DNA was amplified using a thermal cycler (T100TM Thermal Cycler) after 2 min of denaturation at 94°C followed by 55 cycles (30s, 94°C; 45 s, 54°C; 1 min, 72°C). Successful PCRs were selected on ethidium bromide–stained agarose gels. Sanger sequencing was performed in both directions by Eurofins (http://www.eurofins.fr) using M13 tail primers M13F (−21) 5′-TGTAAACGACGGCCAGT-3′; M13R (−27) 5′-CAGGAAACAGCTATGAC-3′ (Messing, 1983).

2.4 | Phylogeographic analysis

Data processing and sequence assembling were performed with Geneious Prime 2020.2.4 (https://www.geneious.com). A total of 91 and 30 sequences, respectively, for *M. brachyurus* and *M. retzii* were obtained for this study. All sequences were aligned with Multiple Alignement using Fast Fourier Transform (Katoh et al., 2002), and percentages of divergence between sequences were calculated using uncorrected p-distances with Geneious Prime 2020.2.4. Genetic diversity of the partial *COI* fragment was examined for both species using Arlequin version 3.11 (Excoffier et al., 2005). For each geographic region, genetic diversity indices were estimated by calculating the number of haplotypes (h), haplotype diversity (H_d), and nucleotide diversity (π).

Median-joining networks were built for *M. brachyurus* and *M. retzii* using Network version 4.6 (Bandelt et al., 1999). We applied the maximum parsimony algorithm and the criterion "frequency >1" to simplify the complex branching scheme and generate networks representing the most parsimonious relationships. To study the partitioning of genetic variance within and among populations, AMOVA were performed using Arlequin v.3.11. The different *M. brachyurus* and *M. retzii* populations were partitioned into large geographical regions, including Central Pacific Ocean (CPO) and West Pacific Ocean (WPO) for *M. brachyurus*, and North Pacific Ocean (NPO), West Indonesia (WI), and East Indonesia (EI) for *M. retzii* due to the small number of individuals for certain localities (Table 1; Figure 1).

The significance of the mitochondrial genetic structures among the different geographic units was then assessed by Φ-statistics (Excoffier et al., 1992) as implemented in Arlequin. Φ_{st} was calculated between each pair of populations. Significance of Φ-statistic was calculated from 10,000 replicates of randomly sampled haplotypes and a α-value of 0.05. Finally, the tests of Fu (Fu, 1997) and Tajima (Tajima, 1983) were performed to see if there was an excess of recent mutations in the sequences, which would correspond to non-neutral processes such as positive selection or population growth.

2.5 | Genetic species delimitation and phylogenetic inferences

Several methods for species delimitation based on DNA sequences have been proposed (Kapli et al., 2017; Pons et al., 2006; Puillandre et al., 2021; Ratnasingham & Hebert, 2013). Each of these methods has different properties, particularly when dealing with singletons (i.e.,

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TABLE 1 Sampling details for *Microphis brachyurus* and *Microphis retzii*, including countries, islands, and phylogeographic area used in the population structure analyses.

Species	Tag	Catalog Number	GenBank ID BOLD BINS	Country	Island	Population
M. retzii	14953	MNHN-IC-2021-0332	OQ221650.1	Indonesia	Ceram	WEST INDONESIA
M. retzii	BIF5345	–	BOLD:ADO3837	Indonesia	Ceram	WEST INDONESIA
M. retzii	14969	MNHN-IC-2024-0058	PQ352220	Indonesia	Ambon	WEST INDONESIA
M. retzii	14973	MNHN-IC-2021-0333	OQ221649.1	Indonesia	Ambon	WEST INDONESIA
M. retzii	14974	MNHN-IC-2021-0334	OQ220383.1	Indonesia	Ambon	WEST INDONESIA
M. retzii	BIF10261	–	BOLD:ADO3837	Indonesia	Sulawesi	WEST INDONESIA
M. retzii	BIF10266	–	BOLD:ADO3837	Indonesia	Sulawesi	WEST INDONESIA
M. retzii	BIF3193	MZB-BIF-3193	BOLD:ADO3837	Indonesia	Papua	WEST INDONESIA
M. retzii	BIF3198	MZB-BIF-3198	BOLD:ADO3837	Indonesia	Papua	WEST INDONESIA
M. retzii	BIF3199	MZB-BIF-3199	BOLD:ADO3837	Indonesia	Papua	WEST INDONESIA
M. retzii	BIF1939	MZB-BIF-1939	BOLD:ACT8563	Indonesia	Java	EAST INDONESIA
M. retzii	–	–	KU692624.1	Indonesia	Java	EAST INDONESIA
M. retzii	BIF1940	MZB-BIF-1940	KU692625.1	Indonesia	Java	EAST INDONESIA
M. retzii	BIF1942	MZB-BIF-1942	KU692628.1	Indonesia	Java	EAST INDONESIA
M. retzii	BIF2809	MZB-BIF-2809	KU692621.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2811	MZB-BIF-2811	KU692622.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2812	MZB-BIF-2812	KU692626.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2814	MZB-BIF-2814	KU692627.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2815	MZB-BIF-2815	KU692629.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2817	MZB-BIF-2817	KU692623.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF2818	MZB-BIF-2818	KU692630.1	Indonesia	Bali	EAST INDONESIA
M. retzii	BIF3775	–	BOLD:ACT8563	Indonesia	Lombok	EAST INDONESIA
M. retzii	BIF3776	–	BOLD:ACT8563	Indonesia	Lombok	EAST INDONESIA
M. retzii	BIF4073	–	PQ352218	Indonesia	Lombok	EAST INDONESIA
M. retzii	BIF4074	–	PQ352219	Indonesia	Lombok	EAST INDONESIA
M. retzii	–	–	KT355109	Taiwan	–	NPO
M. retzii	–	–	KT355110	Taiwan	–	NPO
M. retzii	–	–	KT355111	Taiwan	–	NPO
M. retzii	–	–	KT355112	China	–	NPO
M. retzii	–	–	KT355113	China	–	NPO
M. brachyurus	17761	MNHN-IC-2024-0059	PQ352221	PNG	New britain	WPO
M. brachyurus	17764	MNHN-IC-2024-0059	PQ352222	PNG	New britain	WPO
M. brachyurus	17763	MNHN-IC-2021-0318	OQ221716.1	PNG	New britain	WPO
M. brachyurus	17765	MNHN-IC-2021-0318	OQ221713.1	PNG	New britain	WPO
M. brachyurus	17766	MNHN-IC-2021-0318	OQ221715.1	PNG	New britain	WPO
M. brachyurus	17767	MNHN-IC-2021-0318	OQ221714.1	PNG	New britain	WPO
M. brachyurus	19190	MNHN-IC-2021-0326	OQ221718.1	Solomon	Kolombangara	WPO
M. brachyurus	18254	MNHN-IC-2024-0060	PQ352250	Solomon	Isabel	WPO
M. brachyurus	18256	MNHN-IC-2023-0049	PQ352251	Solomon	Isabel	WPO
M. brachyurus	ES01	MNHN-IC-2024-0061	PQ352231	Indonesia	Borneo	WPO
M. brachyurus	ES02	MNHN-IC-2024-0061	PQ352232	Indonesia	Borneo	WPO
M. brachyurus	ES03	MNHN-IC-2024-0061	PQ352223	Indonesia	Borneo	WPO
M. brachyurus	JP31	MNHN-IC-2024-0062	PQ352239	Japan	Okinawa	WPO
M. brachyurus	JP32	MNHN-IC-2023-0275	PQ352225	Japan	Okinawa	WPO
M. brachyurus	JP33	MNHN-IC-2023-0275	PQ352226	Japan	Okinawa	WPO

(Continues)

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TABLE 1 (Continued)

Species	Tag	Catalog Number	GenBank ID BOLD BINS	Country	Island	Population
M. brachyurus	JP34	MNHN-IC-2024-0063	PQ352224	Japan	Okinawa	WPO
M. brachyurus	JP35	MNHN-IC-2023-0274	PQ352228	Japan	Okinawa	WPO
M. brachyurus	JP36	MNHN-IC-2024-0062	PQ352227	Japan	Okinawa	WPO
M. brachyurus	JP37	MNHN-IC-2024-0062	PQ352240	Japan	Okinawa	WPO
M. brachyurus	JP38	MNHN-IC-2024-0062	PQ352229	Japan	Okinawa	WPO
M. brachyurus	JP39	MNHN-IC-2024-0062	PQ352230	Japan	Okinawa	WPO
M. brachyurus	JP40	MNHN-IC-2024-0062	PQ352233	Japan	Okinawa	WPO
M. brachyurus	JP41	MNHN-IC-2021-0319	OQ221720.1	Japan	Okinawa	WPO
M. brachyurus	JP42	MNHN-IC-2021-0319	OQ221719.1	Japan	Okinawa	WPO
M. brachyurus	JP43	MNHN-IC-2021-0320	OQ221717.1	Japan	Okinawa	WPO
M. brachyurus	JP44	MNHN-IC-2021-0320	OQ221712.1	Japan	Okinawa	WPO
M. brachyurus	JP45	MNHN-IC-2024-0063	PQ352236	Japan	Okinawa	WPO
M. brachyurus	JP46	MNHN-IC-2024-0063	PQ352234	Japan	Okinawa	WPO
M. brachyurus	JP47	MNHN-IC-2024-0063	PQ352237	Japan	Okinawa	WPO
M. brachyurus	JP48	MNHN-IC-2023-0276	PQ352235	Japan	Okinawa	WPO
M. brachyurus	JP49	MNHN-IC-2024-0063	PQ352241	Japan	Okinawa	WPO
M. brachyurus	PFV01	MNHN-IC-2024-0064	PQ352252	FP	Tahiti	CPO
M. brachyurus	PFV02	MNHN-IC-2024-0064	PQ352260	FP	Tahiti	CPO
M. brachyurus	PFV03	MNHN-IC-2024-0065	PQ352265	FP	Tahiti	CPO
M. brachyurus	PFV04	MNHN-IC-2024-0065	PQ352284	FP	Tahiti	CPO
M. brachyurus	PFV05	MNHN-IC-2024-0065	PQ352253	FP	Tahiti	CPO
M. brachyurus	PFV07	MNHN-IC-2024-0065	PQ352261	FP	Tahiti	CPO
M. brachyurus	PFV08	MNHN-IC-2024-0065	PQ352287	FP	Tahiti	CPO
M. brachyurus	PFV11	MNHN-IC-2024-0066	PQ352291	FP	Tahiti	CPO
M. brachyurus	PFV12	MNHN-IC-2021-0323	OQ221725.1	FP	Tahiti	CPO
M. brachyurus	PFV13	MNHN-IC-2024-0067	PQ352280	FP	Tahiti	CPO
M. brachyurus	PFV14	MNHN-IC-2024-0067	PQ352277	FP	Tahiti	CPO
M. brachyurus	PFV15	MNHN-IC-2024-0067	PQ352288	FP	Tahiti	CPO
M. brachyurus	PFV16	MNHN-IC-2024-0068	PQ352290	FP	Tahiti	CPO
M. brachyurus	PFV17	MNHN-IC-2024-0068	PQ352254	FP	Tahiti	CPO
M. brachyurus	PFV18	MNHN-IC-2024-0068	PQ352285	FP	Tahiti	CPO
M. brachyurus	PFV19	MNHN-IC-2024-0069	PQ352286	FP	Tahiti	CPO
M. brachyurus	PFV21	MNHN-IC-2024-0069	PQ352292	FP	Tahiti	CPO
M. brachyurus	PFV22	MNHN-IC-2024-0070	PQ352278	FP	Tahiti	CPO
M. brachyurus	PFV23	MNHN-IC-2024-0070	PQ352266	FP	Tahiti	CPO
M. brachyurus	PFV24	MNHN-IC-2024-0070	PQ352279	FP	Tahiti	CPO
M. brachyurus	PFV25	MNHN-IC-2024-0071	PQ352255	FP	Tahiti	CPO
M. brachyurus	PFV26	MNHN-IC-2024-0072	PQ352259	FP	Tahiti	CPO
M. brachyurus	PFV27	MNHN-IC-2024-0072	PQ352267	FP	Tahiti	CPO
M. brachyurus	PFV28	MNHN-IC-2024-0073	PQ352256	FP	Tahiti	CPO
M. brachyurus	PFV29	MNHN-IC-2024-0074	PQ352281	FP	Tahiti	CPO
M. brachyurus	PFV30	MNHN-IC-2023-0075	PQ352269	FP	Tahiti	CPO
M. brachyurus	PFV32	MNHN-IC-2021-0324	OQ221723.1	FP	Tahiti	CPO
M. brachyurus	PFV33	MNHN-IC-2021-0324	OQ221726.1	FP	Tahiti	CPO
M. brachyurus	PFV34	MNHN-IC-2021-0325	OQ221724.1	FP	Tahiti	CPO
M. brachyurus	PFV35	MNHN-IC-2021-0325	OQ221722.1	FP	Tahiti	CPO

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TABLE 1 (Continued)

Species	Tag	Catalog Number	GenBank ID BOLD BINS	Country	Island	Population
M. brachyurus	PFV36	MNHN-IC-2021-0325	OQ221721.1	FP	Tahiti	CPO
M. brachyurus	PFV37	MNHN-IC-2024-0076	PQ352263	FP	Tahiti	CPO
M. brachyurus	PFV38	MNHN-IC-2024-0076	PQ352274	FP	Tahiti	CPO
M. brachyurus	PFV39	MNHN-IC-2023-0052	PQ352282	FP	Tahiti	CPO
M. brachyurus	PFV40	MNHN-IC-2024-0077	PQ352257	FP	Tahiti	CPO
M. brachyurus	PFV41	MNHN-IC-2024-0078	PQ352289	FP	Tahiti	CPO
M. brachyurus	PFV42	MNHN-IC-2023-0053	PQ352270	FP	Tahiti	CPO
M. brachyurus	PFV43	MNHN-IC-2024-0079	PQ352258	FP	Tahiti	CPO
M. brachyurus	PFV44	MNHN-IC-2024-0080	PQ352271	FP	Tahiti	CPO
M. brachyurus	PFV45	MNHN-IC-2024-0080	PQ352283	FP	Tahiti	CPO
M. brachyurus	PFV46	MNHN-IC-2024-0081	PQ352272	FP	Tahiti	CPO
M. brachyurus	PFV47	MNHN-IC-2024-0082	PQ352275	FP	Tahiti	CPO
M. brachyurus	PFV48	MNHN-IC-2024-0083	PQ352276	FP	Tahiti	CPO
M. brachyurus	PFV49	MNHN-IC-2024-0083	PQ352264	FP	Tahiti	CPO
M. brachyurus	PFV50	MNHN-IC-2024-0084	PQ352268	FP	Tahiti	CPO
M. brachyurus	PFV52	MNHN-IC-2023-0050	PQ352273	FP	Tahiti	CPO
M. brachyurus	PFV53	MNHN-IC-2024-0085	PQ352261	FP	Tahiti	CPO
M. brachyurus	RTNC033	MNHN-IC-2023-0273	PQ352246	NC	–	WPO
M. brachyurus	RTNC109	MNHN-IC-2023-0272	PQ352242	NC	–	WPO
M. brachyurus	RTNC116_A	MNHN-IC-2024-0087	PQ352245	NC	–	WPO
M. brachyurus	RTNC116_B	MNHN-IC-2024-0087	PQ352247	NC	–	WPO
M. brachyurus	RTNC117_A	MNHN-IC-2023-0272	PQ352248	NC	–	WPO
M. brachyurus	RTNC117_B	MNHN-IC-2023-0272	PQ352238	NC	–	WPO
M. brachyurus	RTNC117_C	MNHN-IC-2023-0272	PQ352243	NC	–	WPO
M. brachyurus	RTNC135	MNHN-IC-2021-0322	OQ221709.1	NC	–	WPO
M. brachyurus	RTNC136	MNHN-IC-2021-0322	OQ221711.1	NC	–	WPO
M. brachyurus	RTNC139	MNHN-IC-2021-0322	OQ221708.1	NC	–	WPO
M. brachyurus	RTNC140	MNHN-IC-2024-0088	PQ352249	NC	–	WPO
M. brachyurus	RTNC145	MNHN-IC-2023-0277	PQ352244	NC	–	WPO

Abbreviations: CPO, Central Pacific Ocean; FP, French Polynesia; NC, New Caledonia; NPO, North Pacific Ocean; PNG, Papua New Guinea; WPO, West Pacific Ocean.

lineages represented by a single species) or heterogeneous speciation rates among lineages (Luo et al., 2018). A combination of different approaches is increasingly used to overcome potential pitfalls arising from uneven sampling (Kekkonen et al., 2015; Kekkonen & Hebert, 2014; Sholihah et al., 2020). We used five different sequence-based methods of species delimitation to identify MOTUs: (1) assemble species by automated partitioning (ASAP) (Puillandre et al., 2021), available at https://bioinfo.mnhn.fr/abi/public/asap/; (2) Poisson tree process (PTP) in its single (sPTP) and multiple rates versions (mPTP), as implemented in the stand-alone software mtp_0.2.3 (Kapli et al., 2017; Zhang et al., 2013); (3) general mixed Yule-coalescent (GMYC) in its single (sGMYC) and multiple threshold version (mGMYC), as implemented in the R package Splits 1.0–19 (Fujisawa & Barraclough, 2013). Both the mPTP algorithm and the GMYC use phylogenetic trees as input file. We reconstructed a maximum likelihood (ML) tree for the former using IQ-TREE (Nguyen et al., 2015) with the most likely substitution model according to ModelFinder following the Bayesian information criterion (BIC) (Kalyaanamoorthy et al., 2017), available at http://iqtree.cibiv.univie.ac.at (Trifinopoulos et al., 2016). For the GMYC algorithm, we calculated an ultrametric, fully resolved tree using the Bayesian approach implemented in BEAST 2.6.2 (Bouckaert et al., 2014). Sequences were collapsed into haplotypes prior to reconstructing the ultrametric tree using the ALTER online portal (www.sing-group.org/ALTER/), and Bayesian reconstructions were based on a strict clock of 1.2% of genetic divergence per million years (Bermingham et al., 1997), with a yule model and uniform birth rate tree priors. Two Markov chains of 20 million each were run independently using a GTR + I + Γ substitution models. Trees were sampled every 5000 states after an initial burn-in period of 5 million. Both runs were examined using Tracer 1.7.1 (Rambaut et al., 2018) (effective sample size >200), combined using LogCombiner 2.6.2, the maximum credibility tree, median age

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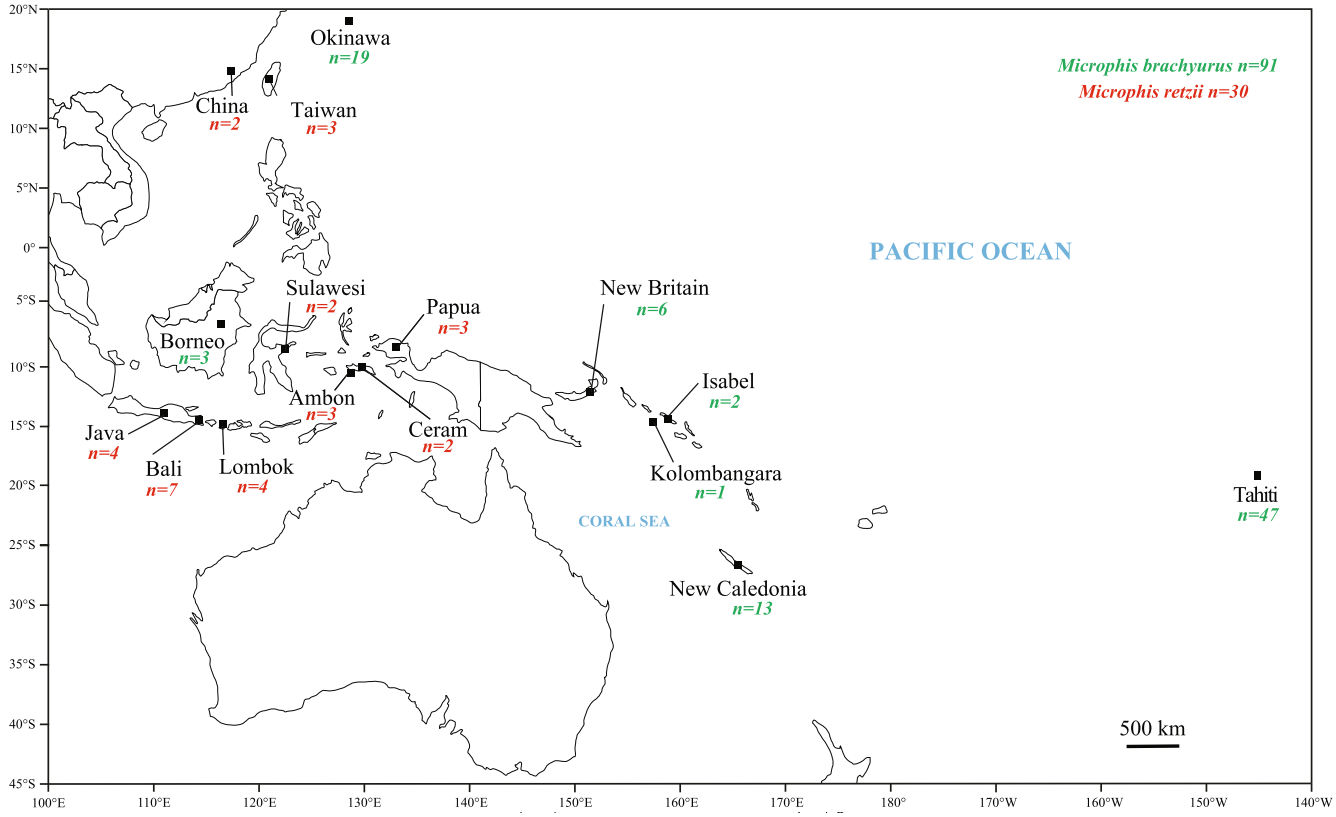

FIGURE 1 Sampling localities. *Microphis brachyurus* are presented in green and *Microphis retzii* are presented in red.

estimates, and corresponding 95% highest posterior density (HPD) were constructed using TreeAnnotator 2.6.2 (Bouckaert et al., 2014). Once obtained, results of the five delimitation species were combined, and a final consensus scheme was established based on a majority-rule consensus. The maximum credibility tree is presented for a visual examination of MOTU divergence and boundaries for both species.

3 | RESULTS

3.1 | M. brachyurus

3.1.1 | Phylogeography

A total of 91 sequences (480 bp) were successfully generated and aligned. Ten haplotypes (H1–H10) were identified, including seven unique haplotypes observed in a single individual (H2, H3, H4, H5, H7, H8, and H9) (Figure 2; Table S1).

The haplotype network for *M. brachyurus* has two main haplogroups separated by five nucleotide substitutions (Figure 2a). Haplogroup 1, composed of haplotypes H1–H9, includes individuals from Papua New Guinea (New Britain), Indonesia (Borneo), Solomon Islands (Kolombangara and Isabel), Japan (Okinawa), and New Caledonia. Haplogroup 1 presents a star-like topology, with a central haplotype (H1) shared by several individuals from most sampling sites and radiating into eight haplotypes separated by a single or two mutations from H1. Haplogroup 2 is composed of one unique haplotype H10 carried by individuals from French Polynesia (Tahiti) only. There are no shared haplotypes between the two haplogroups.

The different sampling sites of *M. brachyurus* were grouped into two large geographical areas to match the spatially nested design of the AMOVA: WPO and CPO (Figure 2b), corresponding to the two haplogroups mentioned earlier. H_d and π are higher in the WPO zone ($H_d = 0.667$; $\pi = 0.927$) compared to the CPO zone ($H_d = 0.117$; $\pi = 0.117$). The values of Fu's F and Tajima's D are significantly negative for the WPO region (Table 2).

The spatial genetic structuring of *M. brachyurus* observed in Figure 2 is supported by a high and statistically significant Φ_{st} value (0.914) between CPO and WPO zones ($p < 0.05$). The average genetic distance between and within CPO and WPO was calculated from the uncorrected p-distances matrix with a value of 1.26% between regions and low genetic distances on average within regions, that is, 0.17%–0% (Table 3).

3.1.2 | MOTUs delimitation and phylogenetic analysis

DNA-based species delimitation methods resulted in congruent delimitation schemes with two MOTUs for sPTP and ASAP, and one MOTU for mPTP and sGMYC (Figure 3, Table S2). mGMYC were not

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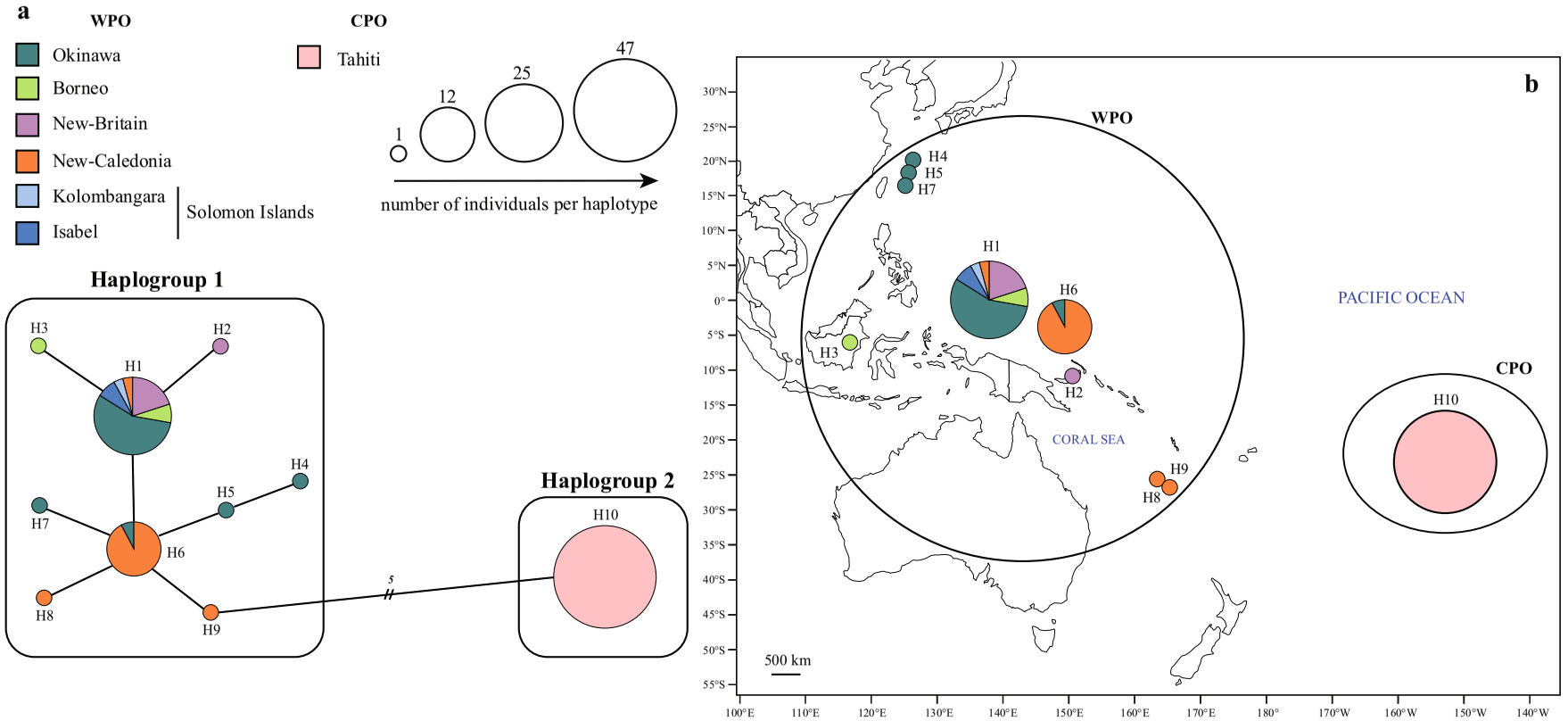

FIGURE 2 (a) Median-joining network for *Microphis brachyurus*. Circles are sized proportionally to the haplotype frequencies. Number of mutations between haplotypes is indicated in italics (it is equal to 1 if no number is indicated). (b) Indo-Pacific map showing the haplotype network and the two geographical areas, Central Pacific Ocean (CPO) and West Pacific Ocean (WPO), for *M. brachyurus*.

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TABLE 2 Molecular diversity indices for *Microphis brachyurus*.

Geographical areas	*N*	F_{st}	H_d	h	π	Fu's F	Tajima's D
WPO	44	0.913	0.667	2	0.927	**−6.083**	**−1.557**
CPO	47	0.916	0.117	10	0.117	−0.521	−0.791

Note: Significant values (*p*-value <0.05) are indicated in bold.
Abbreviations: CPO, Central Pacific Ocean; F and D, neutrality tests; Fst, intra-zone differentiation parameter; Hd, haplotype diversity; h, number of haplotypes; π, nucleotide diversity; *N*, number of individuals sampled; WPO, West Pacific Ocean.

TABLE 3 Pair-wise distance matrix presenting the average divergence percentages for the partial cytochrome oxidase I (*COI*) gene between the two geographical areas (West Pacific Ocean [WPO] and Central Pacific Ocean [CPO]) for *Microphis brachyurus*.

	WPO	CPO
WPO	**0.17**	
CPO	1.26	**0**

Note: Intra-zone divergences are presented in bold.

available for this dataset; the Markov chain failed to run for the multiple threshold version of GMYC with splits. The final consensus scheme consisted of two MOTUs in the Pacific area: one for the West Pacific (Indonesia, Japan, New Caledonia, Papua New Guinea, and Solomon Islands) and one for the Central Pacific (French Polynesia). These two MOTUs were assigned to the same nominal species, *M. brachyurus* (Bleeker, 1854), according to the low genetic divergence (1.26%) (Table 3). The Bayesian gene tree, based on the MOTUs recognized here, suggests a recent diversification of the *M. brachyurus* mitochondrial lineages (Figure 3). Among the 10 haplotypes, 2 lineages are recognized within *M. brachyurus*, one is restricted to the Central Pacific (H10 in French Polynesia) and one is shared in the West Pacific area (H1–H9 from Japan to New Caledonia) with a most recent common ancestor (MRCA) dated around 0.43 MYA (95% HPD: 0.1754–0.759) (Figure 3).

3.2 | M. retzii

3.2.1 | Phylogeography

An alignment of a total of 30 sequences (605 bp) was obtained. Fourteen haplotypes (H1–H14) were identified, including seven unique haplotypes represented by a single individual H1, H4, H5, H6, H8, H11, and H12 (Figure 4; Table S1).

The haplotype network of *M. retzii* showed three distinct haplogroups separated from each other by several mutations and reconstructed haplotypes (Figure 4a). Haplogroups 1 and 2 are separated from each other by 23 mutations and one hypothetical haplotype. Haplogroups 1 and 3 are separated from each other by 16 mutations and one hypothetical haplotype. Haplogroups 2 and 3 are separated from each other by 39 mutations and two hypothetical haplotypes. Haplogroup 1, composed of haplotypes H1–H5, includes individuals from Maluku islands (Ambon and Ceram), Sulawesi, and Papua. Haplogroup 2 is composed of haplotypes H6–H12, which includes individuals from the Sunda Shelf (Bali and Java) and Lesser Sunda islands (Lombok). Finally, haplogroup 3, composed of haplotypes H13 and H14, includes individuals from the North Pacific (Taiwan and China).

The different localities sampled for *M. retzii* were partitioned in three large geographical areas, including NPO, WI (Maluku Islands, Sulawesi, and Papua), EI (Sunda Shelf) (Figure 3b). H_d is highest for EI (0.8) and lowest for NPO (0.6). π is highest for NPO (3) and lowest for WI (0.836) (Table 4). Fu's F and Tajima's D tests were not significant (Table 4).

The spatial genetic structuring of *M. retzii* observed in Figure 3 is supported by high and statistically significant ($p < 0.05$) Φ_{st} values between each zone (Table 5). The highest Φ_{st} value (0.964) is observed between WI and EI, and the lowest Φ_{st} value (0.906) is observed between EI and NPO. The percentages of divergence between the populations of the three zones vary between 3.3% and 5.1% (*x* genetic distance), with shallow divergence among populations ranging from 0.2% to 0.4% (Table 6).

3.2.2 | MOTUs delimitation and phylogenetic analysis

DNA-based species delimitation methods resulted in congruent delimitation schemes with three MOTUs for mPTP, sGMYC, and ASAP and four MOTUs for sPTP and mGMYC (Figure 5; Table S2). The final consensus scheme consisted of three MOTUs in Southeast Asia: one for EI (Ceram, Ambon, Sulawesi, Papua), one for WI (Lombok, Bali, Java), and one for the NPO (China, Taiwan). These three MOTUs were assigned to one nominal species *M. retzii* (Bleeker, 1856). However, the main pair-wise divergences between these MOTUs are relatively high, between 3.3% and 5.1% (Table 6). The Bayesian gene tree, based on the MOTUs recognized here, suggests a diversification of *M. retzii* MOTUs around 1.8 MYA (Figure 5). Among the 14 haplotypes recognized within *M. retzii*, three lineages are observed: one is restricted to EI (H1–H5) with an MRCA dated around 0.42 MYA (95% HPD: 0.166–0.7217); one is restricted to the NPO (H13 and H14) with a MRCA dated around 0.27 MYA (95% HPD: 0.0503–0.5411); and one is restricted to WI (H6–H12) with an MRCA dated around 0.33 MYA (95% HPD: 0.1237–0.5862) (Figure 5).

4 | DISCUSSION

The present study on *M. brachyurus* and *M. retzii* revealed deep genetic divergence of their mitochondrial lineages across the Pacific, with several distinct MOTUs with non-overlapping geographic

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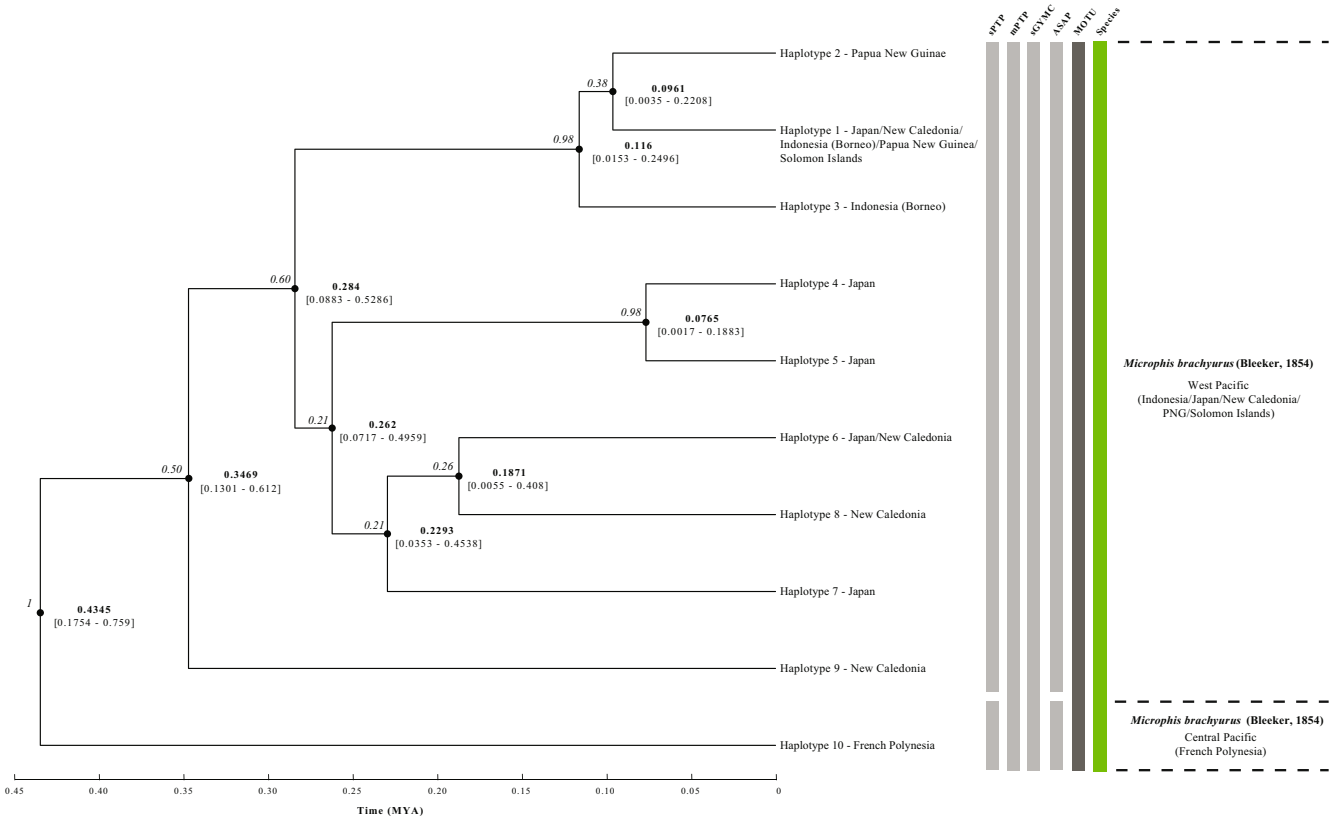

FIGURE 3 Bayesian tree inferred with BEAST 2.6.2 based on *Microphis brachyurus* haplotypes with a GTR + I + Γ substitution model, a lognormal clock model 1.2% of genetic divergence per million years, including 95%, and a coalescent model, including posterior probabilities >0.95 (in italics), highest posterior density (HPD) interval for node age estimates (in brackets), age estimation (in bold), genetic species delimitation results for the four methods and their 50% consensus, and species names.

distributions. By aggregating 91 and 30 sequences for *M. brachyurus* and *M. retzii*, respectively, from various localities, we were able to highlight different evolutionary histories in these two species, which can be explained by a combination of biotic and abiotic factors. These results will have implications for their taxonomy and conservation.

4.1 | *M. brachyurus*, a widespread species

M. brachyurus haplotype network revealed two distinct mitochondrial haplogroups: one in the CPO (French Polynesia) and the other in the WPO (Figure 2a,b). There are no haplotypes shared by individuals from these two areas, suggesting reduced connectivity among maternal lineages of the French Polynesia and the WPO, a hypothesis confirmed by the high and significant Φ_{st} value between these two areas (0.914), showing a deep genetic structuring. The isolation of the mitochondrial lineages of the French Polynesian populations was previously observed in other widespread amphidromous species in the Indo-Pacific, such as the fish species *Sicyopterus lagocephalus* (Pallas, 1770) (Lord et al., 2012) and *Eleotris fusca* (Forster, 1798) (Mennesson et al., 2018) or the prawn *Macrobrachium lar* (Fabricius, 1798) (Castelin et al., 2013). The duration of the marine phase in teleosts has traditionally been used as a proxy of species' dispersal abilities; a species with high dispersal capacities (i.e., long marine phase) tends to have a wider geographic range than those with low dispersal potential (Kinlan & Gaines, 2003). For instance, *Stenogobius genivittatus* (Valenciennes, 1937) widely distributed in the Indo-Pacific, *Awaous guamensis* (Valenciennes, 1937) largely distributed over the Pacific Ocean, or *S. lagocephalus* (Pallas, 1770) widely distributed in the Indo-Pacific exhibit marine phase durations of 135 ± 9.2 days (Radtke et al., 1988), 161 ± 5.7 days (Radtke et al., 1988), and 199 ± 33 days (Hoareau et al., 2007), respectively. In the case of *M. brachyurus*, this duration was estimated to be 38.6 ± 13.1 days (Haÿ et al., 2023b) and may be considered relatively short in comparison to other amphidromous species in the Indo-Pacific region. However, the duration of this marine phase is not always an accurate proxy of their dispersal capacity, which can vary between species with the marine stage of similar duration. For instance, *Sicyopterus sarasini* (De Beaufort, 1915) spends 76.9 ± 3.9 days at sea, but it is endemic to New Caledonia (Lord et al., 2010). Along the same line, *Sicyopterus japonicus* (Tanaka, 1909) is endemic to Taiwan and southern Japan despite a very long marine phase duration of 163.7 ± 12.8 days (Shen & Tzeng, 2008). However, *S. japonicus* is the only temperate species of Sicydiinae goby, its life cycle is also controlled by seasonality as opposed to tropical species. Therefore, abiotic factors can also explain this isolation of the Polynesian mitochondrial lineage from the rest of the Pacific. The different

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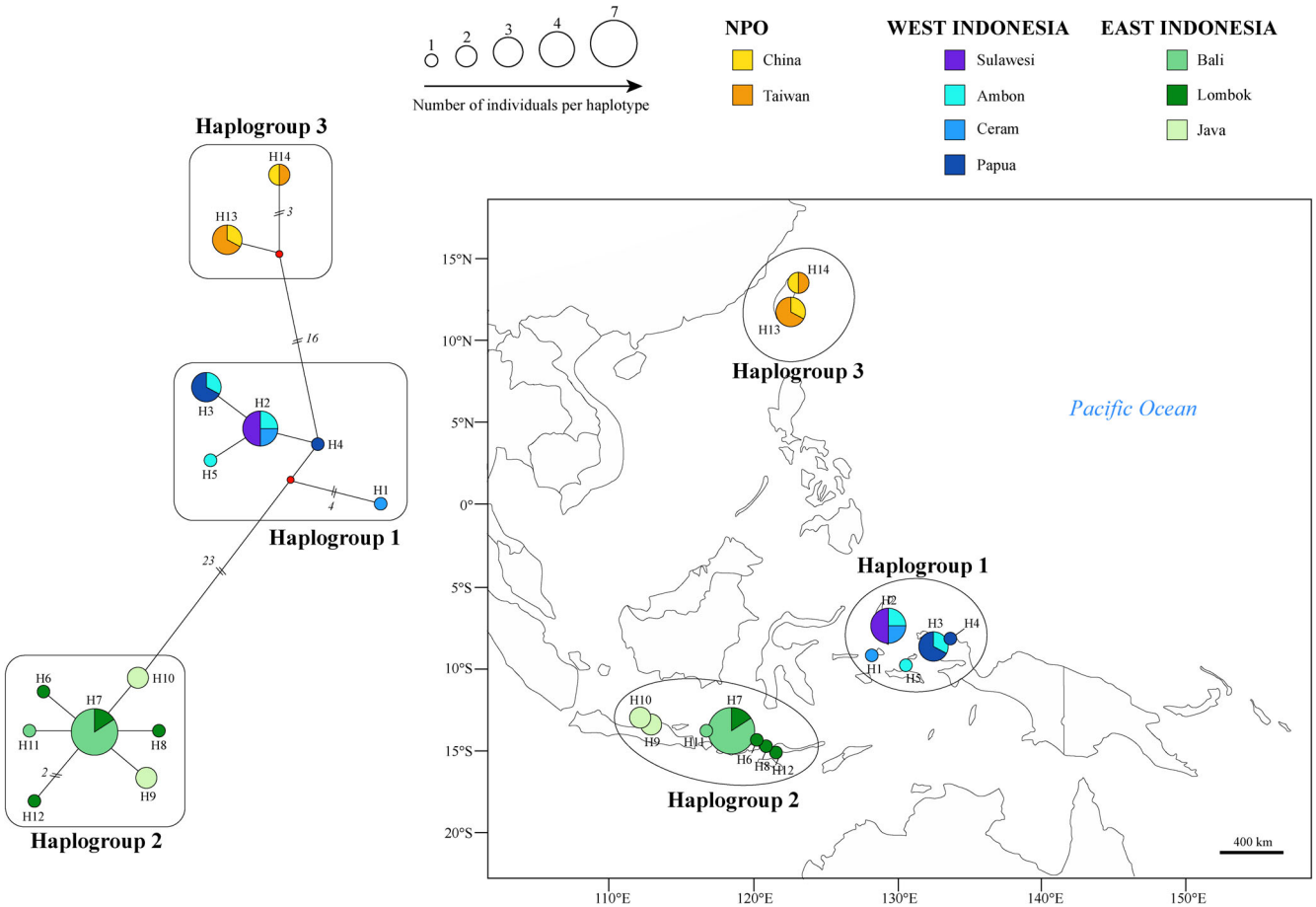

FIGURE 4 (a) Median-joining network for *Microphis retzii*. Circles are sized proportionally to the haplotype frequencies. Number of mutations between haplotypes is indicated in italics (it is equal to 1 if no number is indicated). (b) Indo-Pacific map showing the haplotype network and the three geographical areas: West Indonesia (haplogroup 2), East Indonesia (haplogroup 1), and North Pacific Ocean (NPO; haplogroup 3).

currents encountered by planktonic individuals during the marine dispersal phase may explain this particular distribution of populations (Abdou et al., 2015). Indeed, the Southern Equatorial Current (SEC), the Southern Equatorial Countercurrent (SECC), and the Marquesas Countercurrent (MCC) disperse organisms in different ways; SEC tends to transport individuals sporadically from the Polynesian zone to other West Pacific islands, whereas SECC and MCC tend to limit this transport (Gaither et al., 2010) (Figure 6). Seabed topography can also act as a barrier to dispersal between these two geographical areas (Planes & Fauvelot, 2002). The presence of the Tonga and Kermadec trenches in northern New Zealand over 10,000 m deep could limit the movement of larvae. The combined action of these different factors, biotic (short marine phase, migratory behavior of individuals at sea) and abiotic (bathymetry, large geographic distance between EPO and WPO) would therefore have caused this lack of connectivity between populations of *M. brachyurus* from the WPO and EPO (French Polynesia), a trend previously observed among coral reef fishes (Hubert et al., 2017).

This isolation of the mitochondrial lineages reflects a relatively recent event of colonization of French Polynesia, as the most recent common ancestor between WPO and CPO haplotypes is dated around 0.3 MYA (Figure 3). Indeed, there is only an average of 1% of genetic divergence between these two populations for the partial *COI*. This genetic divergence is below the 3% threshold defined by some authors (Dettaï et al., 2011) to consider two different species based on the *COI* barcode fragment. Therefore, we consider *M. brachyurus* as a unique species in the Pacific zone, with distinct populations in West and Central Pacific. However, if the geographic isolation of French Polynesia persists, these divergences may lead to geographic isolation and evolutionary divergence between these two sets of populations. Dawson (1979) considered *M. brachyurus* as a species complex for which the global distribution could allow it to be divided into four subspecies: *M. brachyurus brachyurus* distributed from Sumatra to the Society Islands; *M. brachyurus millepunctatus* distributed from Sri Lanka to the coasts of East Africa; *M. brachyurus aculeatus* distributed from the coasts of West Africa; and *M. brachyurus lineatus* distributed from the Caribbean islands on the western coasts of South America from Panama to Brazil. Although no morphological differences were identified by Dawson between these different species/subspecies, deep genetic divergences (> 20%) were observed within this complex (Stiller et al., 2022). A large-scale phylogeographic study would then allow to highlight phylogeographic patterns, infer

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TABLE 4 Molecular diversity indices for *Microphis retzii*.

Geographical areas	*N*	F_{st}	H_d	h	π	Fu's F	Tajima's D
NPO	5	**0.941**	0.6	2	3	3.526	1.685
East Indonesia	15	**0.949**	0.8	7	1143	−0.216	−0.301
West Indonesia	10	**0.951**	0.69	14	0,836	−0.116	−0.627

Note: Significant values (*p*-value <0.05) are indicated in bold.
Abbreviations: *N*, number of individuals sampled; NPO, North Pacific Ocean; Fst, intra-zone differentiation parameter; H_d, haplotype diversity; h, number of haplotypes; π, nucleotide divers; F and D, neutrality tests.

TABLE 5 Pair-wise Fst values estimated between the three biogeographic zones for *Microphis retzii*.

	NPO	East Indonesia	West Indonesia
NPO			
East Indonesia	**0.906**		
West Indonesia	**0.952**	**0.964**	

Note: Significant values (*p*-value <0.05) are presented in bold.
Abbreviation: NPO, North Pacific Ocean.

TABLE 6 Pair-wise distance matrix presenting the average divergence percentages for the partial cytochrome oxidase I (*COI*) gene between the three geographical areas (North Pacific Ocean [NPO], West Indonesia, and East Indonesia) for *Microphis retzii*.

	East Indonesia	West Indonesia	NPO
East Indonesia	**0.4**		
West Indonesia	4.44	**0.2**	
NPO	3.3	5.1	**0.4**

Note: Intra-zone divergences are presented in bold.

connectivity of these different groups on a global scale, and define more precisely their taxonomic status. We therefore corroborate Dawson's (1979) findings.

4.2 | *M. retzii*, a species complex?

The haplotype network obtained for *M. retzii* revealed three distinct haplogroups: one in EI (Ceram/Ambon/Papua/Sulawesi, haplogroup 1), one in WI (Bali/Java/Lombok, haplogroup 2), and one in the NPO (China and Taiwan, haplogroup 3) (Figure 4), and no haplotypes are shared between these three haplogroups. This phylogeographic pattern suggests at least distinct mitochondrial divergences among these sets of populations as a result of limited connectivity between these regions. It is currently unknown if *M. retzii* is amphidromous. Haÿ et al. (2023b) have validated an amphidromous life cycle for *Microphis nicoleae*, a closely related species to *M. retzii*, with a relatively short marine phase of 19.7 ± 5.8 days. However, it is important to note that the life cycle of a taxon is not fixed and can vary. The loss of amphidromy is quite common in fish species or populations (Liao et al., 2020; Murase & Iguchi, 2019) and has already been observed in freshwater pipefish (Lord et al., 2024). The marked genetic structuring of these three lineages could be partly explained by biotic factors, such as life-cycle variations (i.e., facultative amphidromy or short marine duration), which limit dispersal and enhance geographic isolation.

These three lineages are found in different areas in Southeast Asia, which is divided into several biogeographic subregions (or hotspots), of which three are represented here: the Sunda Shelf (represented by haplogroup 2), Wallacea (represented by haplogroup 1), and Philippines (represented by haplogroup 3) (Woodruff, 2010) (Figure 4). These subregions present complex biogeographical histories, leading to major vicariance events (Hutama et al., 2016; Lohman et al., 2011). Indeed, genetic structuring can be significantly impacted by biogeography and environmental conditions; rapid changes during the geological history of a region can create barriers to dispersal, which in turn limits gene flow (Lohman et al., 2011; Sholihah et al., 2021a; Sholihah et al., 2021b; Wibowo et al., 2023). Several events on the scale of geological time have caused successive interruptions of connectivity in Southeast Asia. For example, sea-level fluctuations during the glacial cycles of the Pleistocene (2.7 MAY–11,700 years) have led to the establishment of geographical barrier by connecting Borneo, Sumatra, and Java to the mainland, a process that happened repeatedly during the late Pleistocene (Sholihah et al., 2021a; Sholihah et al., 2021b; Woodruff, 2010). Moreover, the Makassar Strait (known as Wallace's line), between the Sunda Shelf and Wallacea, although known to serve as a marine barrier to the dispersal of land animals to Borneo and Sulawesi, could be involved as a dispersal barrier to marine organisms and therefore lead to the genetic isolation of amphidromous species. For instance, sharp genetic breaks were described for populations of the mantis shrimp *Haptosquilla pulchella* among these oceanographic regions, suggesting that Wallace's line has a role in shaping species distribution and population structure (Barber et al., 2000). Murphy and Austin (2005) also suggested a possible effect of Wallace's line on *Macrobrachium rosenbergii*, a freshwater prawn, for which strong genetic divergences are observed between the Australian and Thai populations. Therefore, oceanic currents in Southeast Asia could be involved in the isolation of lineages on each side of Wallace's lines like the Indonesian Throughflow current passing in the Makassar Strait (Godfrey, 1996), thus reducing the connectivity between these two subregions. Isolation of the NPO population (Taiwan and China) can also be influenced by a combination of currents present in this area. The presence of the Kuroshio Current (Figure 6), on the western side of the NPO basin, could act as a dispersal barrier and promote lineage diversification or population

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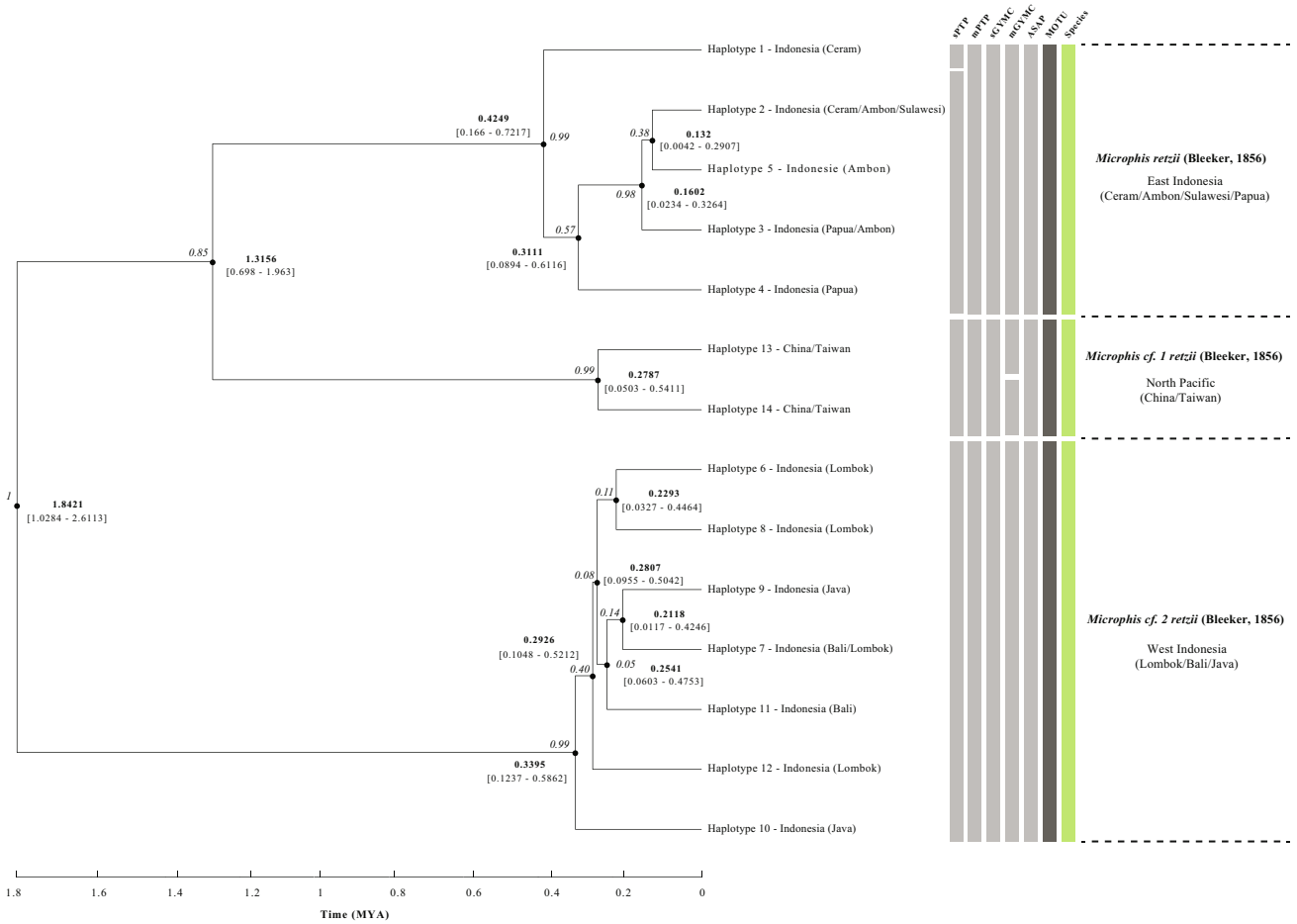

FIGURE 5 Bayesian tree inferred with BEAST 2.6.2 based on *Microphis retzii* haplotypes with a GTR + I + Γ substitution model, a lognormal clock model 1.2% of genetic divergence per million years, including 95%, and a coalescent model, including posterior probabilities >0.95 (in italics), highest posterior density (HPD) interval for node age estimates (in brackets), age estimation (in bold), genetic species delimitation results for the five methods and their 50% consensus, and species names.

differentiation, as it has been observed in some marine organisms and the gobioid *Periophtalmus modestus* (He et al., 2015). Iida et al. (2010) have also shown the important role of the Kuroshio Current to maintain the population structure in the amphidromous goby *S. japonicus* from Taiwan to northern Japan, thus limiting the range of this species to the islands of the North Pacific. The isolation of the different lineages of *M. retzii* may have been influenced by these past and current barriers.

Considering our results, what is the taxonomic status of these different lineages? Compared to *M. brachyurus*, mitochondrial MRCA for *M. retzii* lineages is relatively more ancient and was dated around 1.8 MYA (Figure 5). However, these divergence age estimates should be considered with caution, as the existence of discontinuous gene flow and divergent mitochondrial lineages within the complex may violate the assumptions of the phylogenetic reconstructions of haplotypes presented here (i.e., the lack of genetic structuring). The percentages of divergence between individuals from the three geographical areas are relatively high, ranging from 3.3% between NPO (China and Taiwan) and WI to 5.1% between WI and EI (Table 6), with high and significant Φ_{st} values (Table 5). These results suggest that the three mitochondrial haplogroups of *M. retzii* represent closely related species as follow: (i) *M. retzii* (Bleeker, 1856) (type locality: Manado, Sulawesi, Indonesia) present in Sulawesi, Ceram, Ambon, and in Papua (haplogroup 1) (ii) *M.* cf.*1 retzii* present in WI on the islands of Java, Bali, and Lombok (haplogroup 2), and (iii) *M.* cf.*2 retzii* present in the North Pacific, China, and Taiwan (haplogroup 3). Each of these mitochondrial lineages is therefore restricted to limited geographical areas. These results were expected as endemism is high in this region (Parenti, 2011). Endemism between various close islands of the Indo-Pacific and Indonesia has already been observed (De Mazancourt et al., 2020; Dwiyanto et al., 2021; Haÿ et al., 2021; Jamonneau et al., 2024; Keith et al., 2015; Lord et al., 2012; Wibowo et al., 2023). Indonesian ichthyofauna hosts several radiations of morphologically similar species, and the use of molecular approaches allows us to uncover hidden diversity, including either cryptic or unnamed taxa (Hubert et al., 2015; Kottelat & Lim, 2021; Sholihah et al., 2021a; Sholihah, Delrieu-Trottin, Sukmono, et al., 2021b; Utami et al., 2022). The species of this complex are morphologically very similar, but their

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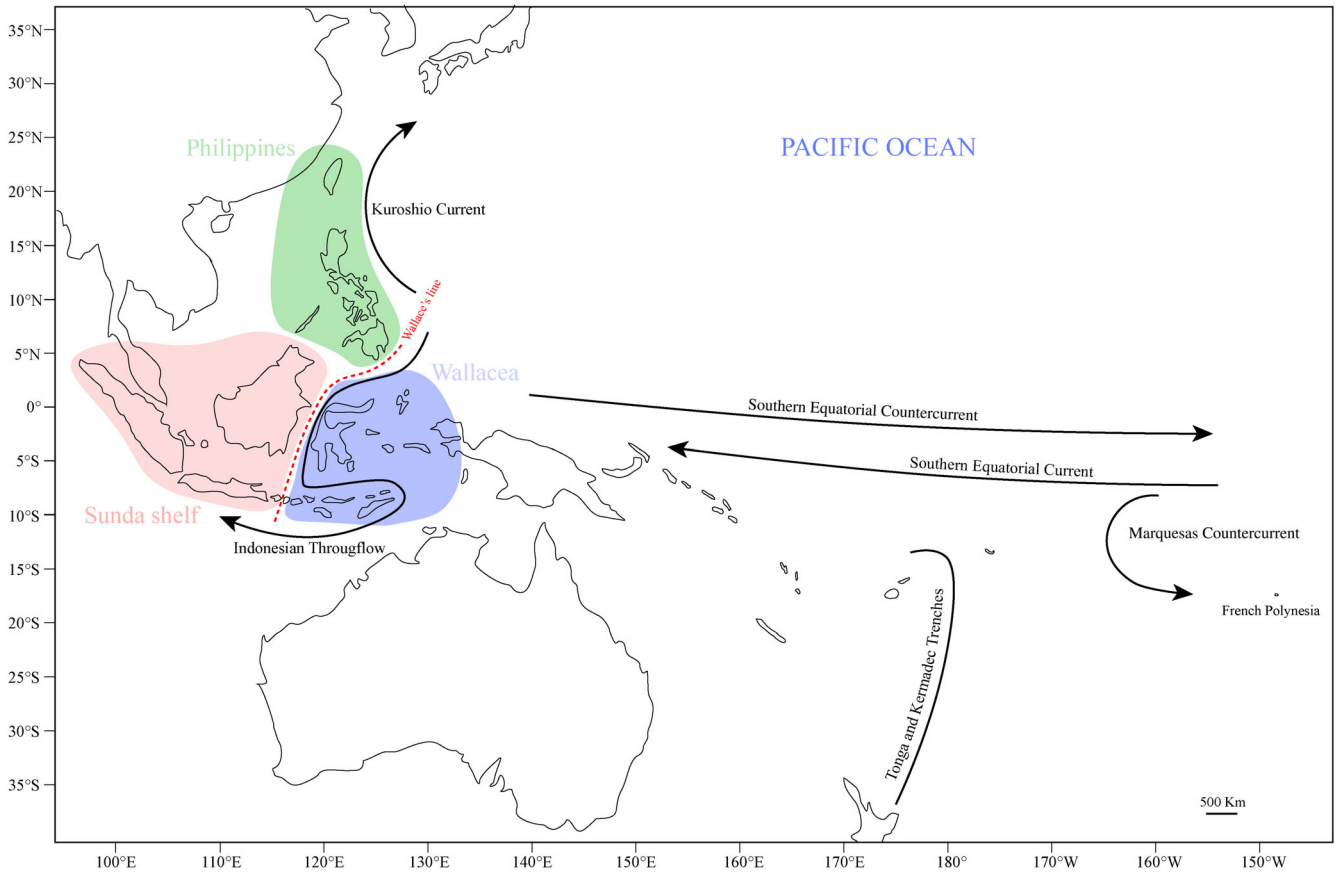

FIGURE 6 Schematic diagram of major currents in the Pacific area with the biogeographic subregions in the Southeast Asia.

differentiation, based on genetics and geography, constitutes strong argument in favor of elevating them to the species level. These results warrant a taxonomic revision of *M. retzii* in Indonesia based on the analysis of nuclear markers, including more specimens and more localities (especially in Borneo, Sulawesi, and the Philippines), and a detailed examination of their morphological characters to revalidate or describe these two potentially new taxa from Java/Bali and from China/Taiwan. This work is in progress.

4.3 | Conservation

Understanding migratory patterns and genetic structuring of teleostean populations is a key element to implement management and conservation measures (Godinho & Kynard, 2009; Lennox et al., 2019; McDowall, 1999). Haplotype networks can be a valuable tool for the conservation biology of amphidromous species, serving as a proxy of life-history traits (Fujita et al., 2016). Alongside, species delimitation analyses proved to be helpful in prioritizing conservation efforts by highlighting deep genetic structures (Hutama et al., 2017; Wibowo et al., 2023). The present study on the mitochondrial genetic structure of *M. brachyurus* and *M. retzii* confirms how molecular studies can help in terms of species conservation. Only a few freshwater pipefishes are referenced in the IUCN Red Lists, and only three have a conservation status, namely *Microphis deocata* (Hamilton, 1822) "Near threatened," *Micropis pleurosticus* (Peters, 1868) "Endangered," and *Microphis insularis* (Hora, 1925) "Vulnerable." The remaining species have been assigned Least Concerned or Data Deficient statuses due to the lack of knowledge, despite freshwater pipefishes being considered to be the endangered species within Syngnathidae, along with seahorses (Pollom et al., 2021). Indeed, currently within the Syngnathidae family, only seahorses (*Hippocampus*) are listed in Appendix II of the Convention on International Trade in Endangered Species for Wild Fauna and Flora (CITES). Concerning *M. brachyurus*, the identification of genetic lineages within its own distribution range would allow them to be defined as distinct units or ESUs, designating populations with conservation action priorities (Fraser & Bernatchez, 2001), particularly for the Polynesian one. This categorization makes it possible to assign a special status to taxa below the nominal level of the species by considering the evolutionary processes of the history of a species (Neal et al., 2018). With the geographical isolation of the Polynesian population of *M. brachyurus*, designation of an ESU in French Polynesia would facilitate conservation programmes specific to this area. For *M. retzii*, we highlighted two new lineages in Indonesia (Bali and Java) and in the North Pacific (China and Taiwan), which should lead to the reassessment of the protection status of this species. Endemism and specific diversity are key concepts in conservation biology, which help define biodiversity hotspots; these specific areas can then be prioritized for conservation efforts (Myers, 1988). Protecting the habitats of these species is also crucial, as the increasing anthropogenic

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pressures on insular aquatic environments negatively impact the life cycle of amphidromous species. Amphidromous species, with their oceanic dispersal, have higher dispersal capacities than strictly freshwater species or species of low vagility (Cook et al., 2009; Crandall et al., 2010). Recruitment of juveniles from the ocean to the rivers is an essential step for the completion of their life cycle. Maintaining this system is crucial for the long-term sustainability of freshwater island systems (Keith, 2003). Maintaining ecological corridors between marine and freshwater biomes is therefore essential to limit habitat fragmentation and biodiversity loss. These conditions are then essential to ensure the sustainability of populations of amphidromous species on a local and regional scale (Abdou et al., 2019).

5 | CONCLUSION

Freshwater pipefish are particularly vulnerable, and there are few studies and very little knowledge on these species. To better protect and implement conservation measures, basic information is needed such as species delimitation and their geographic distribution area. Although understanding how populations are structured within each species is also of major importance, it can be very different based on the lineage. The present study addresses each of these issues by characterizing the phylogeographic patterns of these two species. We uncovered two different patterns and likely distinct evolutionary histories: *M. brachyurus* is one unique species in the Pacific, but there are two groups of populations, one in the WPO and one in French Polynesia, with mitochondrial divergence between them. The isolation of populations between the two areas of the Pacific Ocean suggests the definition of two ESUs. The case of *M. retzii* is very different, as it consists of three closely related species distributed across Asia, suggesting the implementation of different conservation strategies. It should be noted that this study is solely based on mitochondrial data with maternal inheritance. It is possible that further insight could be gained by including nuclear genes in future studies. This could lead to a review of the partitioning of the species in this analysis. However, this hypothesis seems unlikely in view of the geographical isolation of the different lineages highlighted here. This work shows the importance of considering each species individually when developing conservation programmes, because although they belong to the same taxonomic group, each has its own evolutionary history and its own behavior, which will impact population structure and the connectivity. Although the taxonomy of the "*Microphis*" group has recently been clarified (Haÿ et al., 2023a), there is still a substantial effort to be undertaken to better understand their diversity, biology, and ecology.

AUTHOR CONTRIBUTIONS

Vincent Haÿ, Philippe Keith, and Clara Lord conceived the study. Vincent Haÿ, Marion I. Mennesson, Camille Carpentier, Nicolas Hubert, Hadi Dahruddin, Sopian Sauri, Daisy Wowor, and Gino Limmon collected, prepared, and analysed the data. All the authors critically reviewed the manuscript, contributed to the final version, and approved the submitted version.

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